

VMS System

Scenario & Dataflow Reference

Every role-to-role flow in the system, at a glance

For Capstone Project Defense

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A. Issue Reporting & Condition Checks

1 Custodian Submits a Vehicle Issue Report

```
[Custodian] --(submits issue report: vehicle, issue_type, severity, description)-->
[System]
  --(status=Pending · vehicle.condition→Needs Inspection)--> [Admin] (sees it in
Issue Reports queue)
```

Admin can ALSO submit issue reports directly (not just Custodian). Both roles can create one.

2 Admin Creates a Ticket From an Issue Report

```
[Admin] --(clicks "Create Ticket" on a Pending report)--> [System]
  --(issue report status: Pending → In Maintenance · ticket created, status=Open)-->
[Custodian] (assigned to inspect)
```

3 Admin Resolves an Issue Report Directly — No Ticket

```
[Admin / Maintenance Personnel] --(edits issue report, sets status)--> [System]
  --(vehicle status/condition synced to match: Resolved→Available/Good, In
Maintenance→Under Maintenance/Needs Repair)--> done
```

No ticket ever required — a false alarm or a quick fix can be closed out this way. A Custodian may only edit/delete their **own** report, and only while it's still Pending.

4 Custodian Logs a Vehicle Condition Check

```
[Custodian] --(inspects vehicle, submits condition_result)--> [System]
  --(latest record per vehicle = "current"; older ones become history)--> [Admin]
(sees it in Condition Monitoring)
```

B. Ticket Lifecycle — Main Issue → Sub-Issues

5 Admin Creates a Maintenance Ticket

```
[Admin] --(creates ticket: vehicle, title, description, priority, assigns Custodian)--> [System]
--(status=Open · duplicate-title check on this vehicle)--> [Custodian] (notified: inspection assigned)
```

Blocked if this vehicle already has an open ticket with the same (normalized) title — must add as a sub-issue instead.

6 Custodian Inspects the Main Issue — Three Branches

```
[Custodian] --(physically inspects vehicle, submits verdict)--> [System]
```

- **No Issues**: no sub-issues created, ticket jumps straight to Admin confirmation, then Closes
- **Needs Maintenance**: sub-issue(s) created, ticket status → Active, ready for mechanic assignment
- **Not Fit for Service**: sub-issue(s) created, vehicle status auto-changes to Out of Service

Attestation ("I have personally inspected this vehicle") is required before submission.

7 Admin Assigns a Mechanic to a Sub-Issue

```
[Admin] --(selects Mechanic + maintenance_type)--> [System]
--(sub-issue status → Under Repair)--> [Mechanic] (notified: work order assigned)
```

8 Mechanic Logs Repair Work

```
[Mechanic] --(logs repair notes, parts used, cost, photo)--> [System]
--(sub-issue status → For Inspection · verification_assigned_to = the ticket's Custodian)--> [Custodian] (notified: verification required)
```

9 Custodian Verifies the Repair — Functional Test

```
[Custodian] --(operates vehicle, checks each functional-test item, attests)-->
[System]
```

- **All Pass** → **Approved**: sub-issue status → For Confirmation --> **[Admin]** notified (repairs approved)
- **Any Fail** → **Rejected**: sub-issue status → Under Repair (bounced back) --> **[Mechanic]** notified (redo repairs)

Only the Custodian explicitly assigned to *this* sub-issue's verification can act — not just anyone holding the Custodian role (matters when one person is both Custodian and Mechanic).

10 Admin Gives Final Confirmation

```
[Admin] --(reviews verified repair, confirms)--> [System]
--(sub-issue status → Done · linked Issue Report → Resolved · unified Maintenance
Record created)--> [Custodian] (notified: sub-issue confirmed)
```

If this was the last unresolved sub-issue on the ticket, every Admin is also notified: "Ticket Ready to Close."

11 Admin Sends a "Confirmed" Sub-Issue Back (Rework)

```
[Admin] --(rejects at confirmation stage)--> [System]
--(sub-issue status → Under Repair · verification cleared)--> [Mechanic] (notified:
work order sent back)
```

12 Admin Unconfirms a "Done" Sub-Issue (Post-Confirmation Adjustment)

```
[Admin] --(clicks "Unconfirm" on a Done sub-issue, adds notes)--> [System]
--(status → For Confirmation · confirmation cleared)--> [Admin] (re-issues
confirmation after adjusting)
```

Lighter than a full reopen — stays within Admin's own hands, doesn't send it back through Custodian verification.

13 Admin Defers a Sub-Issue (Can't Fix Now)

```
[Admin] --(records reason: no budget, part on back-order, etc.)--> [System]
--(sub-issue status → Deferred · auto-opens a follow-up Issue Report)-->
[Custodian] (notified)
```

Ensures a deferred defect isn't silently forgotten — it leaves a breadcrumb.

14 Mechanic/Custodian Adds a Sub-Issue Mid-Ticket

```
[Mechanic] --(discovers unrelated 2nd problem while repairing)--> [System]
--(new sub-issue, status=Open · ticket progress recalculated)--> [Admin] (assigns
it out via scenario 7)
```

Only the ticket's *own* assigned Custodian or an assigned Mechanic can do this — **never Admin**, even on a combined-role account. Ticket must be Active.

15 Admin Reassigns the Mechanic Mid-Repair

```
[Admin] --(mechanic unavailable – sick, transferred)--> [System]
--(assigned_mechanic swapped · prior work log preserved)--> [New Mechanic]
(notified, sees prior notes) + [Old Mechanic] (notified)
```

16 Admin Closes the Ticket — All Sub-Issues Done

```
[Admin] --(closes, N of N sub-issues Done)--> [System]
--(ticket status → Closed)--> [Custodian] + [every assigned Mechanic] (notified:
ticket closed)
```

17 Admin Closes the Ticket as a Decision (Leftovers Deferred)

```
[Admin] --(closes early, records deferral reason for the rest)--> [System]
--(ticket status → Closed · unresolved sub-issues marked Deferred)--> [Custodian] +
[Mechanics] (notified)
```

18 Admin Reopens a Deleted (Archived) Ticket

```
[Admin] --(recovers an accidentally deleted, not-yet-finished ticket)--> [System]
      --(duplicate-Main-Issue check on this vehicle · ticket restored from snapshot)-->
[Custodian] (notified: ticket reopened)
```

Only works for a "Deleted" archive entry — a permanently "Closed" one is locked and cannot be reopened this way.

C. Vehicle Lifecycle

19 Admin Decommissions a Vehicle

```
[Admin] --(records decommission_reason)--> [System]
```

- **Blocked** if any open (non-Closed/Cancelled) ticket exists on this vehicle
- **Else:** status → Decommissioned, condition → Damaged; excluded from every readiness/coverage calculation; history preserved

20 Admin Recommissions a Decommissioned Vehicle

```
[Admin] --(restores)--> [System]
      --(status → Available, condition → Good · decommission fields cleared)--> back in
the active fleet
```

D. Readiness & Preventive Maintenance

21 Custodian/Admin Performs a Readiness Check

```
[Custodian / Admin] --(physically confirms fuelled, equipped, working – per-checklist)--> [System]
    --(any failed item = NOT ready · otherwise verified ready for 24 hrs)--> reflected instantly in Vehicle Management + Dashboard
```

22 Admin/Mechanic Creates a Preventive Maintenance Schedule

```
[Admin / Mechanic] --(vehicle, type, date, optional recurrence)--> [System]
    --(status=Scheduled · double-booking check: 1 per vehicle per date)--> sits on the calendar until due
```

23 Mechanic Marks a Schedule Done — Recurrence Auto-Regenerates

```
[Mechanic / Admin] --(marks done, logs cost/notes)--> [System]
    --(Maintenance Record created as Completed immediately – no verify/confirm gate · schedule → Completed)--> [System]
    --(if recurring: next occurrence auto-created, collision-checked)--> back on the calendar
```

Deliberately simpler than the ticket pipeline — routine, expected, low-risk work; no independent Custodian verification here.

24 Admin/Mechanic Logs a Standalone Maintenance Record

```
[Admin / Mechanic] --(external vendor repair, no ticket involved)--> [System]
    --(vehicle forced Under Maintenance/Needs Repair · progress_status=Assigned)--> [Custodian] (can verify)
```

25 Custodian Verifies / Admin Confirms a Standalone Record

```
[Custodian] --(inspects, submits verification_result)--> [System]
--(only if Passed)--> [Admin] --(confirms)--> [System] --(progress_status →
Completed)
```

Same Tier-1/Tier-2 checks-and-balances as tickets — Completed can **never** be set by direct edit.

E. User & Access Management

26 Admin Creates a New User (Single or Multi-Role)

```
[Admin] --(name, email, password, one or more roles)--> [System]
--(first role in the list becomes "primary" – drives portal routing)--> [New User]
can log in immediately
```

27 Admin Deactivates a User

```
[Admin] --(deactivates)--> [System]
--(is_active=false · every active session/token revoked immediately)--> user locked
out on next request
```

No check for active ticket/mechanic assignments — deactivating someone mid-repair doesn't auto-reassign their work.

28 Admin Reactivates a User

```
[Admin] --(reactivates)--> [System] --(is_active=true)--> user can log in again
```


29 Any User Changes Their Own Password

```
[Any User] --(submits old + new password)--> [System]
--(verifies old password first)--> password updated
```

Self-service only for password — no self-service profile/photo edit, and no "forgot password" email-reset flow exists.

F. Notification Catalog

Every automatic notification the system fires, and who receives it. All are ticket-lifecycle events — issue-report submission, PM schedule due-dates, and decommission/recommission do **not** currently trigger a notification.

Event	Trigger	Recipient
Inspection assigned	Ticket created	Assigned Custodian
Inspection submitted	Custodian submits inspection	All Admins
Work order assigned	Mechanic assigned to sub-issue	That Mechanic
Work order reassigned	Sub-issue reassigned	New Mechanic + Previous Mechanic
Repairs completed	Mechanic logs repair	Assigned Custodian ("verification required")
Repairs approved	Custodian verifies = Approved	All Admins
Repairs rejected	Custodian verifies = Rejected	Assigned Mechanic
Sub-issue confirmed	Admin confirms Done	Assigned Custodian
Ticket ready to close	Last sub-issue confirmed Done	All Admins
Work order sent back	Admin rejects at confirmation	Assigned Mechanic
Repair reopened for verification	Admin reopens a confirmed sub-issue	Assigned Custodian
Sub-issue deferred	Admin defers	Assigned Custodian
Ticket closed	Admin closes ticket	Assigned Custodian + every assigned Mechanic
Ticket reopened	Admin reopens a deleted ticket	Assigned Custodian

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29 verified role-to-role scenarios across 6 categories, plus a complete notification catalog.